

Shubhangi Agarwal

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SUMMARY

I am a researcher with experience in **data mining**, **machine learning**, and **large-scale graph data** analysis. Currently working on a generative AI initiative for Indian languages with IIT Kanpur. I developed statistical and algorithmic methods for subgraph querying in complex networks, with relevance to problems such as information extraction, recommendation systems, and fraud detection. My postdoctoral work focused on HyGraphs, a unified data model for dynamic graph and time-series data, enabling efficient analysis of evolving large-scale datasets.

EDUCATION

Ph.D. in Computer Science and Engineering

Indian Institute of Technology Kanpur, Uttar Pradesh, India

CGPA: 8.25

2014 - 2023

Bachelor of Technology Computer Engineering

Sardar Vallabhbhai National Institute of Technology, Surat, Gujarat, India

CGPA: 8.71

2010 - 2014

PHD THESIS

Subgraph Matching and Mining in Large Graphs

Supervisor: Arnab Bhattacharya

- Developed algorithms for Approximate Subgraph Matching in both deterministic and probabilistic graphs.
- Proposed a Graph Neural Network model for robust node embeddings with positional information.

PUBLICATIONS

- “FogCast: Probabilistic Medium Range Fog Forecasting using Global Machine Learning Weather Prediction Model”, Prasad Deshpande, **Shubhangi Agarwal**, Kritika Bansal, Arnab Bhattacharya and Shivam Tripathi, Quarterly Journal of the Royal Meteorological Society. Accepted for publication.
- “A Case Study of Cross-Lingual Zero-Shot Generalization for Classical Languages in LLMs”, V S D S Mahesh Akavarapu, Hrishikesh Terdalkar, Pramit Bhattacharyya, **Shubhangi Agarwal**, Vishakha Deulgaonkar, Chaitali Dangarikar, Pralay Manna, and Arnab Bhattacharya, 63rd Annual Meeting of the Association for Computational Linguistics (ACL), 2025, Austria.
- “Towards Hybrid Graphs: Unifying Property Graphs and Time Series”, Mouna Ammar, Christopher Rost, Riccardo Tommasini, **Shubhangi Agarwal**, Angela Bonifati, Petra Selmer, Evgeny Kharlamov and Erhard Rahm, 28th International Conference on Extending Database Technology (EDBT), 2025, Spain.
- “Transforming Time Series into Graphs and Back with HyGraph”, Mouna Ammar, **Shubhangi Agarwal**, Angela Bonifati and Erhard Rahm, Proceedings of the Workshops of the EDBT/ICDT Joint Conference, 2025, Spain.
- “VeNoM: Approximate Subgraph Matching with Enhanced Neighbourhood Structural Information”, **Shubhangi Agarwal**, Sourav Dutta and Arnab Bhattacharya, 7th Joint International Conference on Data Science and Management of Data (CODS-COMAD), 2024, India.
- “VerSaChI: Finding Statistically Significant Subgraph Matches using Chebyshev’s Inequality”, **Shubhangi Agarwal**, Sourav Dutta and Arnab Bhattacharya, Proceedings of the International Conference on Information and Knowledge Management (CIKM), 2021, pages 2812-2816, Australia.
- “GraphReach: Position-Aware Graph Neural Network using Reachability Estimations”, Sunil Nishad, **Shubhangi Agarwal**, Arnab Bhattacharya and Sayan Ranu, Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI), 2021, pages 1527-1533, Canada.
- “ChiSel: Graph Similarity Search using Chi-Squared Statistics in Large Probabilistic Graphs”, **Shubhangi Agarwal**, Sourav Dutta and Arnab Bhattacharya, Proceedings of the International Conference on Very Large Data Bases (VLDB), 2020, pages 1654-1668, Japan.

EXPERIENCE

Project Executive Officer

IIT Kanpur, India

Sep 2025 - Present

- Project: ‘BharatGPT: A Suite of Generative AI Tech for India’.
- Funding Agency: Department of Science and Technology (DST), Government of India

Post-doctoral Researcher <i>LIRIS Research Lab, CNRS, Université Claude Bernard Lyon 1, France</i>	May 2024 - Apr 2025
- Project: 'HyGraph: Querying and Analytics on Hybrid Graphs'. - Funding Agency: French Agence Nationale de la Recherche (ANR) and the Deutsche Forschungsgemeinschaft (DFG; German Research Foundation)	
Senior Student Research Associate <i>IIT Kanpur, India</i>	Aug 2022 - Dec 2023
- Project: 'Short and Long-term Fog Prediction for North Indian Cities'. - Funding Agency: Scheme for Transformational and Advanced Research in Science (STAR), Ministry of Education, Government of India.	
Summer Intern <i>Flipkart Internet Private Limited</i>	Jun 2015 - Jul 2015
Worked in a project to prepare a model for identification of 'fake' listings.	

TEACHING

Instructor <i>Université Claude Bernard Lyon 1, France</i>	Sep 2024 – Dec 2024
- Program: M2 - Data and Intelligence for Smart Systems (DISS) - Course: Data Processing and Analytics (Jointly with others)	
Teaching Assistant <i>IIT Kanpur, India</i>	Aug 2014 - Apr 2021
Graded and evaluated projects for various courses of Computer Science.	
Senior Tutor <i>IIT Kanpur, India</i>	Aug 2017 - Apr 2018, Aug 2019 - Apr 2020
Led teams of strength ~60; Assisted in backend management, paper-setting and grading.	
Teaching Assistant for MOOC <i>NPTEL - Remote</i>	July 2017 - Sep 2017
Crafted objective questions and resolved student queries on Fundamentals of Database Systems.	

ADMINISTRATIVE EXPERIENCES

Program Committee Member	2025 - current
GRADES-NDA Workshop - SIGMOD/ PODS (2025), ICWE (2025, 2026), TGD Workshop - EDBT/ ICDT (2025).	
External Reviewer	2020 - current
WSDM (2024), CIKM (2021, 2022), DASFAA (2022), CoDS-COMAD (2020, 2021), KDD (2021).	

TECHNICAL SKILLS

Languages	: C, C++, Python, Java, R, JavaScript, PHP, Shell scripting
Databases	: MySQL, Neo4j, SQLite
Libraries	: PyTorch, Tensorflow, Scikit-learn, Numpy, Pandas, PySpark, Ollama
Tools	: git, LaTeX, Docker, Weka

REFERENCES

Arnab Bhattacharya

Professor, Indian Institute of Technology Kanpur, India

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Sourav Dutta

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Distinguished Professor, Lyon 1 University, France

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